SOLAR POWERED ROBOT FOR SOLID WASTE COLLECTION ON WATER SURFACES

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ABSTRACT

In response to increasing water pollution from solid waste accumulation, the project proposes a novel solution using robotics and renewable energy. Our project focuses on designing a solar-powered robot capable of autonomously collecting waste on water surfaces. Integrating advanced technologies such as Arduino microcontrollers, Raspberry Pi units, sensors (IR, temperature, gps), a Raspberry Pi camera, and solar panels, the aim is to optimize obstacle detection, waste identification, and autonomous operation. Field tests will evaluate the robot's effectiveness in mitigating pollution and scalability for widespread deployment, contributing to environmental conservation efforts.

Keywords: Temperature sensor, IR sensor, GPS sensor, Raspberry Pi, Arduino.

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INTRODUCTION

- The world's water bodies face an escalating crisis of pollution, with marine debris posing a significant threat to aquatic ecosystems, wildlife, and human health.

- The heart of our project lies the vision of a cleaner and healthier environment, where water surfaces are free from the scourge of pollution, and aquatic ecosystems thrive in their natural state

OBJECTIVES

To enable autonomous operation, allowing the robot to navigate water bodies, detect obstacles, identify waste, and execute collection maneuvers without human intervention

To design and construct a prototype that integrates advanced technologies, including Arduino microcontrollers, Raspberry Pi computing units, and a suite of sensors and actuators

To validate the robot's performance and effectiveness in real-world scenarios

To develop a solar-powered robot capable of efficiently collecting solid waste on water surfaces

LITERATURE SURVEY

TITLE OF THE PAPER WITH AUTHOR NAME	JOURNAL NAME/YEAR OF PUBLICATION	FINDINGS	LIMITATIONS
Title: Development of a Solar-Powered Autonomous Robot for River Cleanup Authors: Kamran Arshad, Muhammad Usman, Saad Akhtar, et al.	Environmental Technology, 2020	The solar-powered autonomous robot effectively navigated river environments and collected debris, demonstrating its potential for river cleanup operations.	Challenges were identified in maneuvering the robot in fast-flowing rivers, leading to limitations in debris collection efficiency.
Title: Solar-Powered Unmanned Surface Vehicle for Coastal Waste Management Authors: Junghwan Kim, Taesun Park, Kwanghoon Kim, et al.	Ocean Engineering, 2019	The solar-powered unmanned surface vehicle autonomously collected waste along coastal areas, significantly reducing cleanup time and costs.	Limited onboard storage capacity constrained the robot's ability to collect large volumes of debris before requiring manual intervention.
Title: Autonomous Marine Debris Collection Using Solar-Powered Drones Authors: Robert Smith, Emily Davis, Jason Lee, et al.	Journal of Field Robotics, 2021	Solar-powered drones efficiently identified and collected marine debris from remote coastal regions, demonstrating versatility and scalability.	Adverse weather conditions, such as high winds and heavy rainfall, hampered drone operations and limited collection efficiency.
Title: Robotic Waste Collector for Urban Water Bodies Authors: Priya Sharma, Rahul Verma, Ankit Gupta, et al.	Waste Management & Research, 2020.	The robotic waste collector effectively removed debris from urban water bodies, improving water quality and enhancing aesthetic appeal.	Limited battery life restricted the operational range of the robot, necessitating frequent recharging and limiting continuous operation.

EXISTING SYSTEM

The existing systems for collecting solid waste from water bodies typically involve manual methods and some semi-automated technologies.

1. Boat-like devices equipped with conveyors and nets to collect floating debris. They usually require human operation and supervision.
2. Trash Booms and Barriers are stationary or floating barriers installed in water bodies to trap floating debris, which is then collected manually or by machines.
3. Some automated and semi-automated USVs are used to collect debris from water surfaces. However, they often rely on non-renewable energy sources and have limitations in storage capacity and operational duration.

PROPOSED SYSTEM

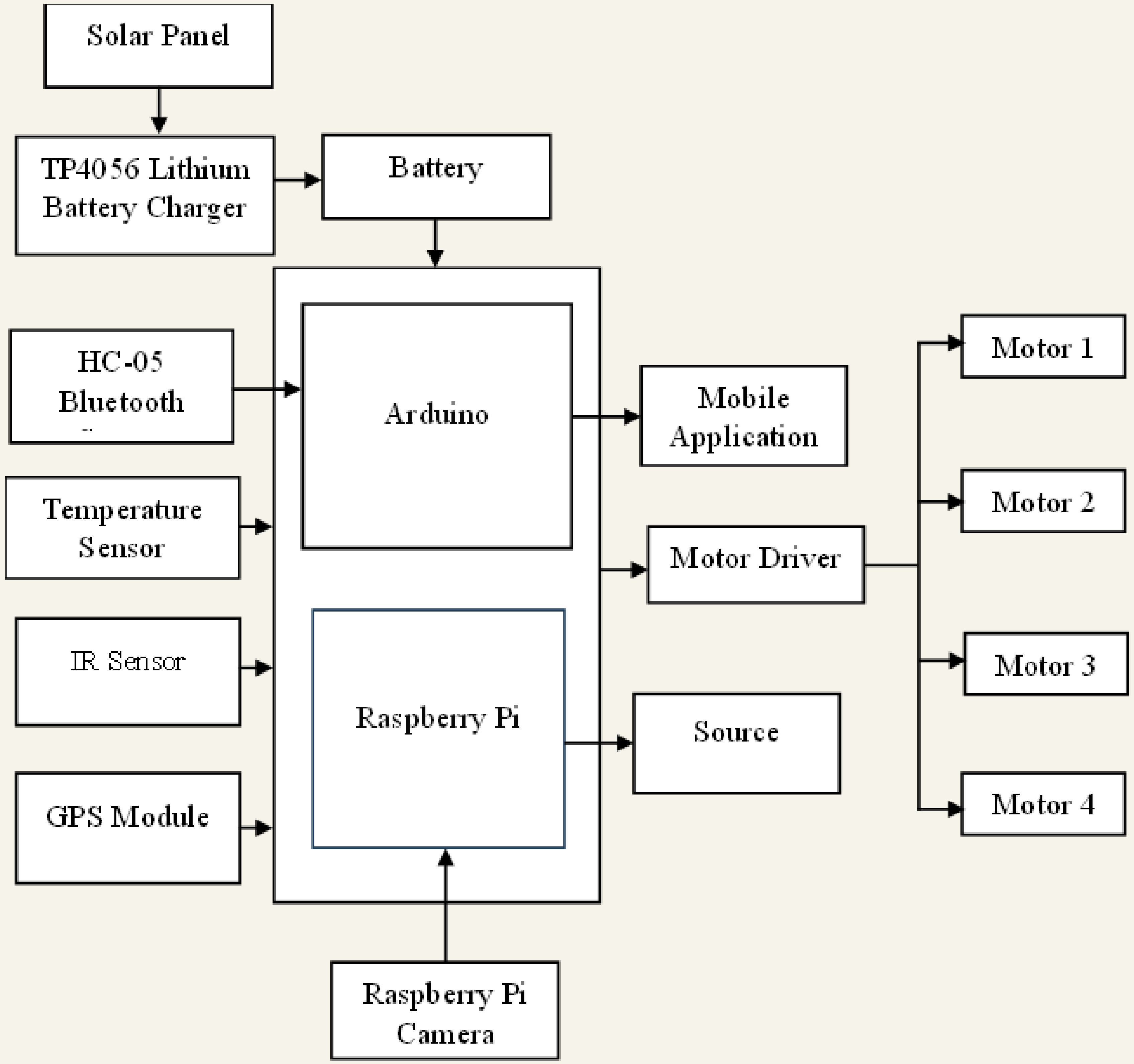
- **Solar Power**
  Utilizes solar panels to power the robot, ensuring sustainable and eco-friendly operation

- **Waste Detection and Collection**
  Uses a combination of cameras and sensors to identify and collect debris effectively

- **Navigation**
  Equipped with advanced sensors and a Raspberry Pi for real-time processing, enabling the robot to navigate water bodies without human intervention.

- **Scalability and Adaptability**
  Designed to be scalable for deployment in various water environments and adaptable to different types of waste and operational conditions.

BLOCK DIAGRAM



Solar Powered Robot for Solid Waste Collection on Water Surfaces

HARDWARE DESCRIPTION

Solar Panel

The solar panel serves as the primary power source for the robot, converting sunlight into electrical energy

Raspberry Pi

Raspberry Pi is a credit-card-sized single-board computer that provides computing capabilities to the robot

Arduino Uno

Arduino Uno is a microcontroller board based on the ATmega328P chip, providing the robot with real-time control over its sensors and actuators

IR Sensor

IR sensor detects infrared radiation emitted or reflected by objects in its field of view.

Bluetooth Sensor

Bluetooth sensor enables wireless communication between the robot and external devices such as smartphones or tablets

Raspberry Pi Camera

Raspberry Pi Camera Module is a high-resolution camera that captures images and videos of the robot's surroundings

FEATURES

The Features are,

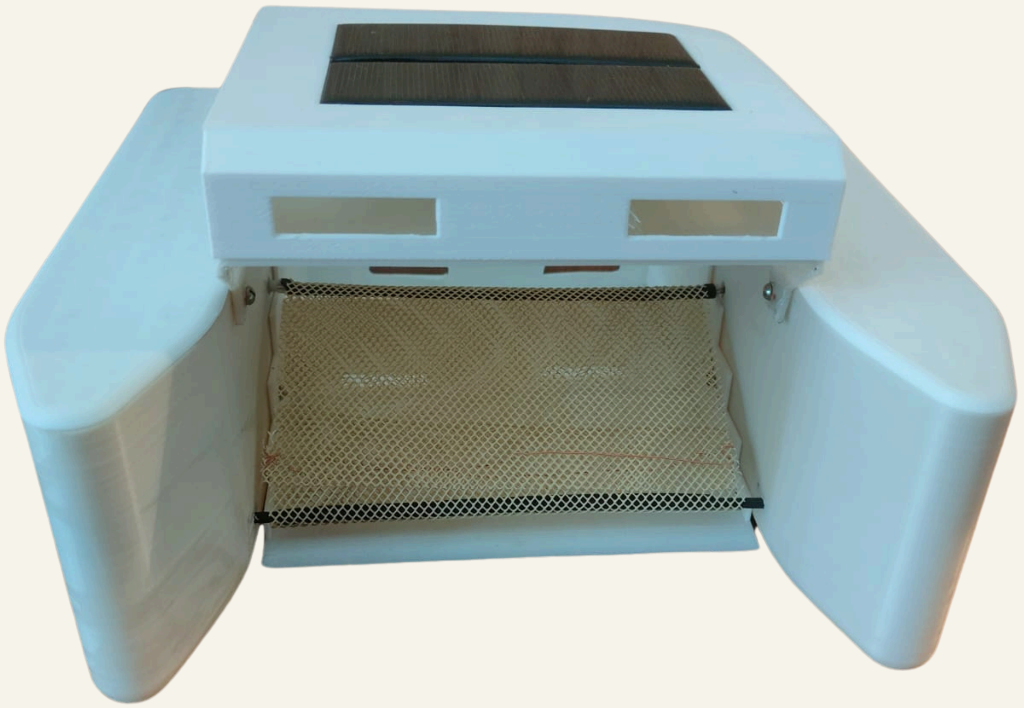
- It harnesses renewable solar energy, ensuring sustainable operation.
- It effectively detects and collects floating debris using integrated cameras and sensors.
- Energy efficiency is optimized through power-saving, extending its operational duration.
- Additionally, remote monitoring and control via Bluetooth facilitate real-time adjustments and data collection.

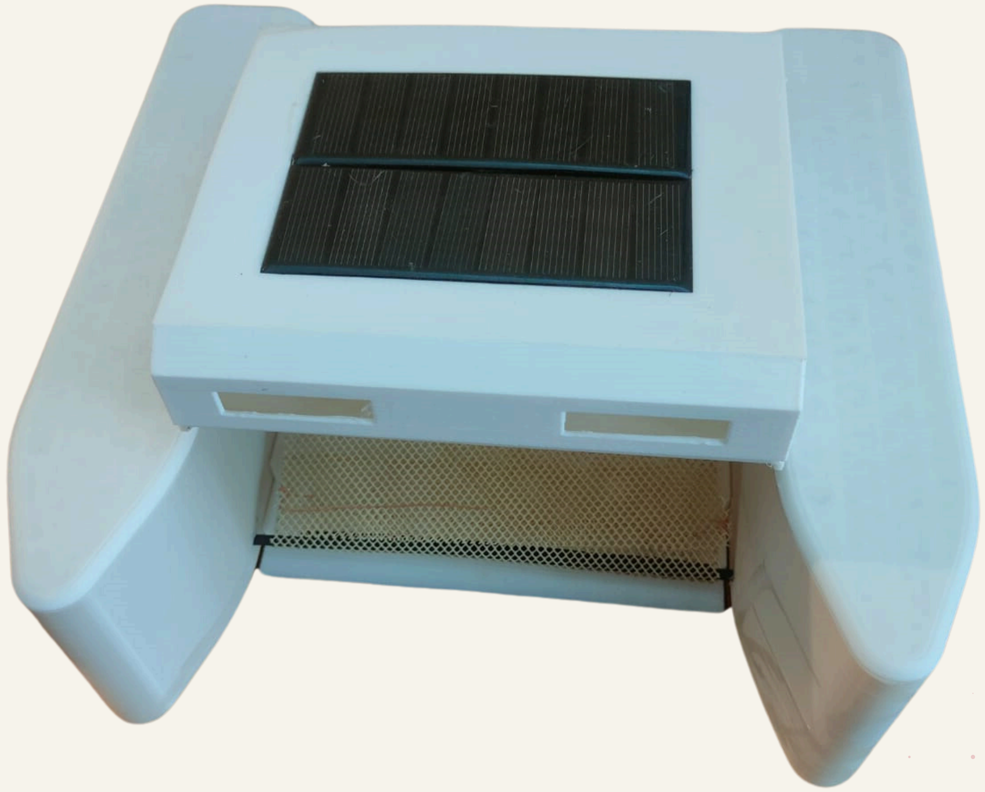
APPLICATION

- The solar-powered robot for solid waste collection has multiple practical applications aimed at improving environmental health and cleanliness.
- It can autonomously clean rivers and lakes, significantly enhancing water quality and preserving aquatic ecosystems. It helps maintain clean and attractive shorelines by efficiently collecting debris.
- Urban water bodies, such as ponds and canals, benefit from reduced pollution and improved aesthetics.

OUTPUT AND RESULT

The solar-powered robot autonomously cleans water bodies like rivers, lakes, and coastal areas, enhancing water quality and aesthetics while providing valuable environmental data through integrated sensors.





CONCLUSION

In conclusion, the development of the solar-powered robot for solid waste collection represents a significant advancement in environmental technology. By harnessing renewable energy and integrating autonomous navigation and waste collection capabilities, the robot offers an efficient and sustainable solution for cleaning water bodies. Its diverse applications, ranging from river and lake cleanup to disaster response and environmental monitoring, underscore its potential impact on improving water quality and ecosystem health. Moving forward, continued refinement and deployment of the robot hold promise for addressing the global challenge of water pollution and advancing environmental conservation efforts.

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THANK YOU

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